Problem

Let $[x]$ denote the greatest integer less than or equal to x.
Find the sum of $[x]$ over all real solutions of

$$16 + 15x + 15x^2 = [x]^3.$$

Source: **IOQM 2024**

§1 Separating the Integer and Fractional Parts

Let

$$x = I + J,$$

where

$$I = [x] \in \mathbb{Z}, \qquad 0 \le J < 1.$$

The equation becomes

$$16 + 15(I + J) + 15(I + J)^2 = I^3.$$

Expanding and rearranging,

$$I^3 - 15I^2 - 15I - 16 = 15J(1 + 2I + J).$$

The cubic on the left factors as

$$I^3 - 15I^2 - 15I - 16 = (I - 16)(I^2 + I + 1).$$

Hence

$$(I - 16)(I^2 + I + 1) = 15J(1 + 2I + J).$$

§2 Restricting the Possible Integer Parts

Since

$$0 \le J < 1,$$

the right-hand side is nonnegative.

Also,

$$I^2 + I + 1 > 0$$

for every integer I.

Therefore,

$$(I - 16)(I^2 + I + 1) \geq 0,$$

which implies

$$I \geq 16.$$

Thus no solution can satisfy $I \leq 15$.

§3 The Solution with Integer Part 16

If

$$I = 16,$$

then

$$0 = 15J(33 + J).$$

Since $33 + J > 0$, we must have

$$J = 0.$$

Hence

$$x = 16$$

is a solution.

§4 The Solution with Integer Part 17

If

$$I = 17,$$

the equation becomes

$$307 = 15J(35 + J).$$

Define

$$F(J) = 15J(35 + J) - 307.$$

Then

$$F(0) = -307 < 0,$$

while

$$F(1) = 15 \cdot 36 - 307 = 233 > 0.$$

Since F is continuous and strictly increasing on $[0, 1]$, there exists a unique

$$J \in (0, 1)$$

such that

$$F(J) = 0.$$

Therefore there is exactly one solution with

$$[x] = 17.$$

§5 Excluding All Larger Integer Parts

We now show that no solution satisfies

$$I \geq 18.$$

Since $I = [x]$, this implies

$$x \geq 18.$$

For $x \geq 18$,

$$x^3 > [x]^3.$$

Thus it suffices to prove

$$x^3 > 15x^2 + 15x + 16.$$

Indeed,

$$x^3 - (15x^2 + 15x + 16) = x^2(x - 15) - 15x - 16.$$

For $x \geq 18$,

$$x - 15 \geq 3,$$

so

$$x^2(x - 15) \geq 3x^2.$$

Hence

$$x^3 - (15x^2 + 15x + 16) \geq 3x^2 - 15x - 16.$$

Now

$$3x^2 - 15x - 16 = 3x(x - 5) - 16.$$

Since $x \geq 18$,

$$3x(x - 5) \geq 3 \cdot 18 \cdot 13 = 702,$$

and therefore

$$3x^2 - 15x - 16 > 0.$$

Consequently,

$$x^3 > 15x^2 + 15x + 16.$$

Since

$$[x]^3 = 15x^2 + 15x + 16,$$

this is impossible. Therefore no solution satisfies

$$x \geq 18.$$

§6 The Final Count

We have found exactly two solutions:

$$x = 16,$$

and one solution lying in the interval

$$17 < x < 18.$$

Their integer parts are therefore

$$16 \quad \text{and} \quad 17.$$

Hence the required sum is

$$16 + 17 = 33.$$

$$\boxed{33}$$